

Name: _____ Date: _____ Period: _____

1. For each transformation rule below, identify the type of transformation (vertical translation, horizontal translation, vertical dilation, or horizontal dilation/reflection) and describe its effect on the graph of f .

(a) $g(x) = f(x) - 5$

Type: **vertical translation** Effect: **shift down 5 units**

(b) $g(x) = f(x - 3)$

Type: **horizontal translation** Effect: **shift right 3 units**

(c) $g(x) = \frac{1}{3}f(x)$

Type: **vertical dilation** Effect: **vertical compression by factor $\frac{1}{3}$**

(d) $g(x) = f(-x)$

Type: **horizontal reflection** Effect: **reflect over the y -axis**

2. The table below gives values of f .

x	-2	-1	0	1	2
$f(x)$	8	2	0	2	8

Complete a table of values for each transformed function.

(a) $g(x) = f(x) + 4$

x	-2	-1	0	1	2
$g(x)$	12	6	4	6	12

(b) $h(x) = -f(x)$

x	-2	-1	0	1	2
$h(x)$	-8	-2	0	-2	-8

3. Let f have domain $[1, 5]$ and range $[-3, 6]$. For each transformed function, state the domain and range.

(a) $g(x) = f(x) + 2$ Domain: **$[1, 5]$** Range: **$[-1, 8]$**

(b) $g(x) = f(x + 4)$ Domain: **$[-3, 1]$** Range: **$[-3, 6]$**

(c) $g(x) = 3f(x)$ Domain: **$[1, 5]$** Range: **$[-9, 18]$**

(d) $g(x) = f(x - 1) - 2$ Domain: **$[2, 6]$** Range: **$[-5, 4]$**

4. Describe all transformations applied to f to obtain $g(x) = -3f(x + 2) + 1$. List them in order of application.

Step 1: Horizontal shift left 2 units ($h = 2 > 0$). Step 2: Vertical stretch by factor 3 and reflect over the x -axis ($a = -3$). Step 3: Vertical shift up 1 unit ($k = 1$)......

5. **(AP-Style Multiple Choice)** Let f be a function with domain $[-4, 4]$ and range $[0, 8]$. Which of the following gives the range of $g(x) = -2f(x) + 1$?

- (A) $[-15, 1]$ ← **correct**
- (B) $[-16, 0]$
- (C) $[1, 17]$
- (D) $[-15, -1]$

Answer: **A** Show your reasoning:

Range of f is $[0, 8]$. Multiply by -2 : $[0, 8] \rightarrow [-16, 0]$ (reversed). Add 1 : $[-16 + 1, 0 + 1] = [-15, 1]$

Reflection

What is the trickiest part of horizontal transformations? Why does $f(x + h)$ shift *opposite* to the sign of h ?

Because $f(x + h)$ evaluates f at $x + h$, so the output at input x equals what f outputs at $x + h$. To get $f(0)$, we need $x + h = 0$, so $x = -h$. The graph shifts to where the input gives zero of the inner expression......